Data Sources:
IoT sensors
Databases
APIs
Logs

Data Ingestion:
Batch ingestion
Stream ingestion

Data Preprocessing:
Cleaning
Normalization
Filtering

Processing Engine :
ETL tasks
Feature extraction
Data transformation

Feedback / Adaptation

Auto-Adaptive Decision Module:
Performance monitoring
Resource optimization
Dynamic adaptation

Data Storage :
Data Lake
Data Warehouse

Visualization & Reporting:
Dashboards
Reports
Alerts